

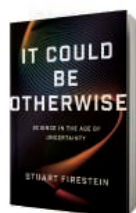
BOOKS ET AL.

SCIENCE AND SOCIETY

The optimism of doubt

Embracing uncertainty, not banishing it, is what makes science powerful

Sorin M. S. Krammer



Stuart Firestein
Basic Books, 2026.
288 pp.

IT COULD BE OTHERWISE: SCIENCE IN THE AGE OF UNCERTAINTY | The title of Stuart Firestein's new book is a disguised provocation. *It Could Be Otherwise* is not a platitude but a hard-won philosophical revolution, one that took most of human history to arrive at and one that we are now, perversely, at risk of forgetting. Spanning the Copernican revolution, Darwinian evolution, chaos theory, and the crisis of public science literacy, the book pursues an ambitious thesis: that science's greatest gift to civilization is not the certainties it has produced but rather the structured, generative relationship with uncertainty it has invented.

Firestein, a cellular neuroscientist whose research operates at the probabilistic frontier, brings personal scientific authority to this argument. To underpin it, he draws on the American pragmatists, the works of Charles Darwin and Ludwig Boltzmann, Ilya Prigogine's thermodynamics, and chaos theory's demonstration that deterministic equations can generate irreducibly unpredictable outcomes. His claim that uncertainty is not a residue of ignorance but rather a permanent feature of reality is more radical than it sounds, but it becomes less contentious as it is corroborated by some of the book's finest chapters.

Its historical sweep is an equally impressive feat. The most evocative episode is Firestein's revisionary reading of Copernicus. Contrary to the familiar demotion narrative, he argues, occupying the center of the medieval cosmos was no honor. It meant inhabiting what historians have called the "ignobilissima," the cosmological drain where all the "crud" of creation collected. Copernicus did not demote us, he promoted us into the heavens, and with that elevation came the radical implication that the laws governing Earth were the laws of the stars.

The chapter audaciously titled "F--- the Enlightenment" is sharper still. Here, Firestein identifies two ideas that Isaac Newton's triumph planted in the scientific culture: a belief in single solutions, which fed directly into eugenics and totalitarian rationalism, and Laplacian determinism, the dream of a "vast intellect" that could predict everything, thus leaving nothing to chance.

Yet this is also where the book's claims of originality mandate scrutiny. David Deutsch's *The Beginning of Infinity* (1) covered much of this ground—the radical openness of an explanatory universe and the link between fallibilism and optimism—with comparable ambition and greater rigor. Prigogine and Isabelle Stengers made the entropy-as-possibility argument four decades ago (2). Firestein synthesizes this tradition fluently but does not sufficiently distinguish his own contribution from it.

He traces the shift from the ancient "what is" to the modern "what if" through five "cogs," his manual for science under uncertainty, including renunciation of authority, experimentation, failure, provisional belief, and counterintuitive thinking. All are pedagogically appealing and well illustrated. The cogs describe how science should work; how it actually works, in the face of paradigm resistance and institutional conservatism, demands a franker reckoning (3) and all the more so in an era of algorithmic automation and autonomous research systems (4).

The cog of provisional belief makes a quietly radical point. Grounded in philosopher Catherine Elgin's notion of being "true enough," it insists that unsettled science is not unsound science but rather science in motion. Yet the framework risks tidying up a history that is considerably messier. The book's own examples undercut it: Boltzmann's probabilistic thermodynamics was ridiculed for decades before vindication, and his suicide is treated as a historical footnote rather than as evidence that science's social dynamics are no more hospitable to productive failure than those of any other human institution.

The book is similarly more confident in diagnosis than in remedy when it comes to public communication. Firestein argues compellingly that schools teach a 19th-century deterministic model of science that leaves students unable to reason under uncertainty, directly feeding the epistemic vulnerability that misinformation exploits, a dynamic well documented in the literature (1, 5). Firestein's proposed solutions, however, curriculum reform and exemplary figures such as Carl Sagan as role models, do not address this adequately.

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There is also a tension that the book never quite resolves: Although uncertainty can be inherently optimistic, it is equally available to bad actors. Eugenics, for example, was deeply probabilistic; the “one-drop rule” weaponized Mendelian statistics to enforce racial segregation. A more critical and prescriptive account of how uncertainty is deployed would have strengthened the book considerably. The final chapter recruits Isaiah Berlin’s liberal pluralism as an ally, but his pluralism was a thesis about human values, not about scientific knowledge. Conscripting him without resolving this tension is a shortcut that philosophers will notice.

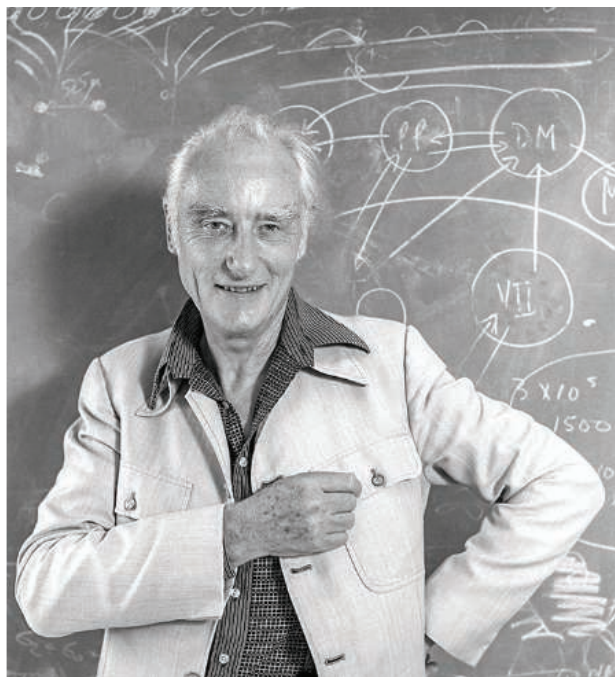
It Could Be Otherwise is, at its core, an act of reframing; uncertainty not as a confession of failure but as the condition of possibility. That reframing is genuinely valuable, and it reaches beyond what either triumphalist or relativist accounts of science can offer. However, where the stakes are highest, namely translating a persuasive philosophical vision into actionable policy, much work remains. □

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PODCAST



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